

AYMANE AIT DADS

AI Engineering & Automation Intern | LLM Workflows · Python Automation · Prompt Engineering
Aymane.Ait-dads@eurecom.fr | +33 7 60 92 50 93 | linkedin.com/in/aymane-ait-dads | Sophia Antipolis, France

PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams in the NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, \$106K+ prize pool) by engineering an end-to-end LoRA SFT pipeline on a 30B-parameter model with BF16 precision and custom verification pipelines. Reduced manual network analysis time by 60% at Orange Maroc by deploying automated feature engineering and clustering pipelines on 100K+ production records, delivering actionable outputs directly to the optimization team. Brings hands-on experience with Python, LLM frameworks (Hugging Face Transformers, LangChain), prompt engineering, API integrations, and agentic pipeline design - skills directly applicable to OMMAX's AI Center of Excellence automation workflows. Aims to contribute to building client-facing PoCs and automated reporting pipelines at OMMAX, combining rigorous ML engineering with practical automation tooling.

CORE COMPETENCIES

Python Automation | Prompt Engineering | API Integration | LLM Frameworks | Automated Reporting Pipelines | Prototype Development | AI Workflow Automation | Data Handling

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built automated feature engineering and model evaluation pipeline on **100K+ fiber-optic network records**, reducing manual analysis time by ~60% and surfacing actionable maintenance recommendations.
- Deployed clustering pipeline mapping customer performance profiles to **5 commercial service tiers** using K-Means; output adopted directly by the network optimization team for production decisions.
- Developed **Random Forest classification models** on upload speed, latency, and jitter signals to classify network quality across thousands of nodes with Scikit-learn and Pandas.
- Delivered stakeholder presentations translating complex model outputs into **business maintenance recommendations**, communicating technical findings to non-technical decision-makers via Power BI dashboards.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge

Kaggle Competition · In Progress (June 2026) · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context) across 5 structured reasoning categories.
- Ran **controlled ablation experiments** across LR schedules, packing strategies, and synthetic augmentation - discovering that reasoning-style consistency outweighs data volume in LLM SFT; built custom cipher parsers and 8-bit operation solvers for verified chain-of-thought data.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM

AI-Generated Image Detection

EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in Class

- Ranked **1st in EURECOM class** (private Kaggle leaderboard, 0.791 AUC) by fine-tuning CLIP ViT-L/14 (428M params) with a custom MLP head and dual-LR strategy (2e-6 backbone, 5e-4 head) on 250K images across 25+ generator types.
- Deployed **3-model ensemble with 10-view TTA** (flips, crops, blur, resizes) weighted by validation AUC; also submitted to NTIRE 2026 @ CVPR CodaBench international benchmark for out-of-distribution generalization evaluation.

PyTorch · OpenCLIP · FP16 Mixed Precision · AdamW · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Cloud Computing, Web Applications, Statistics, Image Security

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

AI & LLM Engineering: Python | LangChain | Hugging Face Transformers | Prompt Engineering | LoRA / PEFT | vLLM | Agentic Pipelines

ML Frameworks & Data: PyTorch | Scikit-learn | Pandas | NumPy | TRL (SFTTrainer) | REST APIs | Supabase / PostgreSQL

Automation & Reporting: Power BI | Matplotlib | Git | Docker (basic) | Linux / Bash | Feature Engineering | RAG